Algebraic and Geometric Representation Theory Program

The 16th Southeastern Lie Theory Workshop

University of Virginia, July 17–21, 2026

Plenary Addresses

All plenary talks in Clark Hall 108

July 17	July 18	July 19	July 20	July 21
Yakimov (9:00–9:50)	Arakawa (9:00–9:50)	Riche (9:00–9:50)	Fintzen (9:00–9:50)	Savage (9:00–9:50)
Coffee Break (10:00–10:30)				
Du (10:30–11:20)	Linshaw (10:30–11:20)	Nakashima (10:30–11:20)	Chan (10:30–11:20)	Wang (10:30–11:20)
Kujawa (11:30–12:20)	Kumar (11:30–12:20)	Informal Discussion (11:30–12:20)	Lai (10:30–12:20)	Informal Discussion (11:30–12:20)
Lunch Break (12:30–14:00)				
Afternoon parallel sessions in Warner Hall				
Parallel Sessions (14:00–16:30)	Parallel Sessions (14:00–16:30)	Free Afternoon	Parallel Sessions (14:00–16:30)	Free Afternoon
Tea and Discussion (16:30–17:30)	Tea and Discussion (16:30–17:30)	Free Afternoon	Tea and Discussion (16:30–17:30)	Free Afternoon

Contributed Talks

Friday, July 17

	Session 1 Warner 110	Session 2 Warner 115
2:00–2:20pm	Drupieski	Sandvik
2:30–2:50pm	Hou	Yu
3:00–3:20pm	Wilbert	Orr
3:30–3:50pm	Zhu	Vu
4:00–4:20pm	Mamani	Jurisich

Saturday, July 18

	Session 1 Warner 110	Session 2 Warner 115
2:00–2:20pm	Grantcharov	Pillen
2:30–2:50pm	Larson	Jing
3:00–3:20pm	Lohan	Maity
3:30–3:50pm	Xie	Upadhyay
4:00–4:20pm	Ngo	Basu

Monday, July 20

	Session 1 Warner 110	Session 2 Warner 115
2:00–2:20pm	Arias	Chen
2:30–2:50pm	Lu	Tang
3:00–3:20pm	Xiang	Vashaw
3:30–3:50pm	Bendel	Zhou
4:00–4:20pm		Wu

Program: Titles and Abstracts of Plenary Addresses

Tomoyuki Arakawa

Affiliation: Okinawa Institute of Science and Technology
Email: arakawa@kurims.kyoto-u.ac.jp
Title: Vertex Superalgebras for Hypertoric Varieties and 3d Abelian Gauge Theories
Abstract: Hypertoric varieties are a class of symplectic singularities and their resolutions, obtained as Hamiltonian reductions of a symplectic vector space acted on by a torus. In physics, they appear as Higgs (and Coulomb) branches of 3d N-4 supersymmetric quantum field theories with abelian gauge group.

In this talk, we construct an $\hbar$-adic sheaf of vertex operator superalgebras over a given smooth hypertoric variety. Its global sections give the A-twisted boundary of the corresponding 3d gauge theory. We use this to prove that the associated affine variety of this hypertoric vertex operator superalgebra recovers the singular hypertoric variety. This proves the 3d Higgs branch conjecture for a large class of boundary vertex operator superalgebras. In particular, these vertex operator superalgebras are quasi-lisse.

This is in contrast to the (purely even) hypertoric vertex operator superalgebras constructed previously by Kuwabara as global sections of sheaves on families of universal Poisson deformations of the hypertoric varieties. These are generally not quasi-lisse. We show that the vertex operator superalgebras defined in this paper are (fermionic) simple-current extensions of those defined by Kuwabara, and investigate the consequences for symplectic duality and characters. We observe that the latter are upgraded from partial (or false) theta functions to quasimodular forms.

This is a joint work with Andrea Ferrari and Sven Möller.

Charlotte Chan

Affiliation: University of Michigan
Email: charchan@umich.edu
Title: Deligne–Lusztig varieties
Abstract: Deligne–Lusztig varieties are natural subvarieties of the flag variety whose cohomology encodes (much of) the representation theory of finite groups of Lie type. In the last two decades, there have been many advances in developing a Deligne–Lusztig theory in thecontext of representations of p-adic groups. I'll discuss recent progress in this subject.

Jie Du

Affiliation: University of New South Wales
Email: j.du@unsw.edu.au
Title: The queer canonical basis theory
Abstract: Canonical basis theory has been established for many quantum/i-quantum groups, quantum linear supergroups and their modified ones. However, the existence of such a theory for quantum queer supergroups $\mathbf{U}_v(\mathfrak{q}_n)$ is still mysterious. This is partly because no appropriate bar involution on $\mathbf{U}_v(\mathfrak{q}_n)$ is known.

A recent completion of a new realization of $\mathbf{U}_v(\mathfrak{q}_n)$ via Hecke-Clifford superalgebras and queer q-Schur superalgebras in [1] gives us a chance for possibly resolving the problem. Building on this

work, we may first establish such a theory for Hecke-Clifford superalgebras and then extend it to $\mathbf{U}_v(\mathfrak{q}_n)$.

In this talk, I will start with bar involutions $\flat$ on Hecke-Clifford superalgebras H_r^c ($r \geq 1$), and explain how to lift them to bar involutions on $\mathbf{U}_v(\mathfrak{q}_n)$. I then introduce a standard basis $\mathcal{B}_r(c)$ for H_r^c and a natural partial ordering $<$ on its index set. Unfortunately, canonical bases associated with the data $(\mathcal{B}_r(c), <, \flat)$ do not exist. However, either replacing $\mathcal{B}_r(c)$ by some new bases or twisting the order $<$ to a new order $<'$ can define canonical bases for H_r^c. Thus, canonical bases associated with $(\mathcal{B}_r(c), <', \flat)$ exist. If time permits, I will mention some possible ways to extend the construction above to queer q-Schur superalgebras and the modified $\mathbf{U}_v(\mathfrak{q}_n)$.

This is based on joint work with H. Gu, J. Wan and Z. Zhou.

Jessica Fintzen

Affiliation: University of Bonn

Email: `fintzen@math.uni-bonn.de`

Title: Reduction to depth-zero for $\bar{\mathbb{Z}}[1/p]$-representations of p-adic groups

Abstract: The category of smooth complex representations of p-adic groups decomposes into Bernstein blocks and by a joint result with Adler, Mishra and Ohara from August 2024 we know that under some minor tameness assumptions each Bernstein block is equivalent to a depth-zero Bernstein block, which are the representations that correspond roughly to representations of finite groups of Lie type. This result allows to reduce a lot of problems about representations of p-adic groups and the Langlands correspondence to their depth-zero counterpart that is often easier to solve or already known.

In this talk we present analogous results for R-representations of p-adic groups where R is any ring that contains all p-power roots of unity, a fourth root of unity and the inverse of a square-root of p, for example, R could be an algebraically closed field of characteristic different from p or the ring $\bar{\mathbb{Z}}[1/p]$. This is a joint work with Jean-François Dat. While the result is analogous to the result with complex coefficients (except for the "blocks" being "larger"), the proof is of a very different nature. In the complex setting the proof is achieved via type theory and an isomorphism of Hecke algebras, which are techniques not available for general R-representations. We sketch in the talk how we deal with the category of R-representations instead.

Jonathan Kujawa

Affiliation: Oregon State Universtiy

Email: `kujawaj@oregonstate.edu`

Title: Interpolating Schur Algebras

Abstract: In this talk, I will describe a one-parameter family of algebras that naturally encompasses the Schur algebras. These algebras are generically semi-simple, they have the ordinary Schur algebras as a quotient, and their modules are naturally a subcategory of a parabolic Category O for the general linear Lie algebra.

This is joint work with Addison Day.

Shrawan Kumar

Affiliation: University of North Carolina, Chapel Hill
Email: shrawan@email.unc.edu
Title: Conjectural Positivity for Pontryagin Product in Equivariant K-theory of Loop Groups
Abstract: Let G be a connected simply-connected simple algebraic group over $\mathbb{C}$ and let T be a maximal torus, $B \supset T$ a Borel subgroup and K a maximal compact subgroup. Then, the product in the (algebraic) based loop group $\Omega(K)$ gives rise to a comultiplication in the topological T-equivariant K-ring $K_T^{\top}(\Omega(K))$. Recall that $\Omega(K)$ is identified with the affine Grassmannian X (of G) and hence we get a comultiplication in $K_T^{\top}(X)$. Dualizing, one gets the Pontryagin product in the T-equivariant K-homology $K_0^T(X)$, which in-turn gets identified with the convolution product (due to S. Kato).

Now, $K_T^{\top}(X)$ has a basis $\{\xi^w\}$ over the representation ring $R(T)$ given by the ideal sheaves corresponding to the finite codimension Schubert varieties X^w in X. We make a positivity conjecture on the comultiplication structure constants in the above basis. Using some results of Kato, this conjecture gives rise to an equivalent conjecture on the positivity of the multiplicative structure constants in T-equivariant quantum K-theory $QK_T(G/B)$ in the Schubert basis.

Chun-Ju Lai

Affiliation: Academia Sinica
Email: cjlai@as.edu.tw
Title: Quantum wreath products and p-adic general linear group
Abstract: The uniqueness of Whittaker models for the p-adic group $G = GL_d(F)$ relies heavily on a deep understanding of the Iwahori component of the Gelfand–Graev G-module. As a module over the affine Hecke algebra, this component identifies with the Kashiwara–Miwa–Stern tensor space at $n = 1$, meaning its corresponding endomorphism algebra is the affine q-Schur algebra $\widehat{S}_q(1, d)$. In this talk, we provide a transparent description of these p-adic objects and their metaplectic (i.e., $n > 1$) analogs via the theory of quantum wreath products. This new perspective leads to results that were previously inaccessible through standard p-adic methods. This is based on joint work with Alexandre Minets (Bonn) and Valentin Buciumas (Pohang).

Andrew Linshaw

Affiliation: University of Denver
Email: andrew.linshaw@du.edu
Title: Dualities of W-algebras
Abstract: W-algebras are a class of vertex algebras that are associated to a Lie (super)algebra g and a nilpotent orbit in the even part of g. Principal W-algebras (i.e., the nilpotent is principal) are the best-studied examples. When g is a Lie algebra, they satisfy Feigin-Frenkel duality, which is the isomorphism between the principal W-algebras of g and its Langlands dual Lie algebra. When g is simply-laced, they satisfy another duality called the coset realization, which was proven in my work with Arakawa and Creutzig in 2018. A common generalization of these dualities was conjectured in 2017 by the physicists Gaiotto and Rapcak and was proven in my work with Creutzig in 2020. I will discuss these dualities and some of their applications in representation theory, as well as some generalizations to W-algebras with supersymmetry in recent work with Creutzig, Kovalchuk, Song, and Suh.

Toshiki Nakashima

Affiliation: Sophia University, Tokyo
Email: toshiki@sophia.ac.jp
Title: Crystal structure of localized quantum unipotent coordinate category and its application
Abstract: We consider the quiver Hecke algebra R associated with a Kac-Moody Lie algebra $\mathfrak{g}$. Let R-gmod denote the category of finite-dimensional graded R-modules, and let C_w be its full subcategory associated with a Weyl group element w. Let $\widetilde{R\text{-gmod}}$ and $\widetilde{C_w}$ be their localizations, referred to as localized quantum unipotent coordinate categories.

It has been shown by M. Kashiwara and T.N. that the set of isomorphism classes of simple objects up to grading shifts, $\mathrm{Irr}(\widetilde{C_w})$, carries a crystal structure and is isomorphic to the cellular crystal $\mathbb{B}_{\mathbf{i}} = B_{i_1} \otimes \cdots \otimes B_{i_k}$, where $\mathbf{i} = i_1 \cdots i_k$ is a reduced word of w.

In the case where $\mathfrak{g}$ is of classical type and w_0 is the longest element of the Weyl group, this isomorphism induces functions ε_i^* on $\mathbb{B}_{\mathbf{i_0}}$, where $\mathbf{i_0}$ is a reduced word of w_0. We provide an explicit description of the functions ε_i^* and, using these, give a characterization of the unit object of $\widetilde{R\text{-gmod}}$.

This is joint work with K. Matsuura.

Simon Riche

Affiliation: Universite Clermont Auvergne
Email: simon.riche@uca.fr
Title: Semiinfinite sheaves on affine flag varieties
Abstract: Semiinfinite sheaves are certain complexes of sheaves on (possibly partial) affine flag varieties of reductive algebraic groups which are equivariant with respect to the "semiinfinite Iwahori subgroup", i.e. the product of the arc group of a maximal torus and the loop group of the unipotent radical of a Borel subgroup. In this talk I'll explain how (infinity-)categories of such complexes can be defined, explain some of their basic properties, and report on connections (partly under construction) with representations of the Langlands dual group (and its Lie algebra).

This is joint work with Pramod Achar, Gurbir Dhillon and Quan Situ.

Alistair Savage

Affiliation: University of Ottawa
Email: alistair.savage@uottawa.ca
Title: Climbing the Temperly-Lieb tower
Abstract: The Temperley-Lieb algebras form a natural tower, where moving up and down corresponds to induction and restriction. In this talk, I will describe a graphical calculus that makes these functors, and the natural transformations between them, visible. The calculus is built from a diagrammatic 2-category with generators for one-step induction and restriction, together with two-step summands coming from cup-cap idempotents. A key result is a basis theorem: morphism spaces have explicit bases indexed by bridges, a family of paths generalizing Dyck paths. This theorem shows that the graphical calculus faithfully captures the corresponding Temperley-Lieb bimodule theory. I will also discuss the decategorified picture, where standard modules are represented by

homogenized Chebyshev polynomials, and idempotent completion leaves the Grothendieck ring unchanged.

Weiqiang Wang

Affiliation: University of Virginia
Email: `ww9c@virginia.edu`
Title: Simple character formula for finite W-algebras of type A
Abstract: We will present a canonical basis character formula for the irreducible modules in parabolic BGG-type categories, including the category of finite-dimensional modules, for finite W-(super)algebras of type A. These categories categorify the tensor product modules of irreducible polynomial representations (and their duals) over a quantum group of type A. Moreover, the standard modules and irreducible modules in these categories categorify the standard basis and Lusztig's dual canonical basis in the tensor product modules.

This is joint work with Shun-Jen Cheng.

Milen Yakimov

Affiliation: Northeastern University
Email: `m.yakimov@northeastern.edu`
Title: Short star products for quantum symmetric pairs and quantum Popov degenerations
Abstract: The goal of the talk is to describe a bridge between two directions in representation theory. The first is the theory of short star products and twisted traces of Etingof-Stryker. The second is the theory of quantum symmetric pairs of Letzter-Kolb based on the classification of Kac-Wang of automorphisms of Kac-Moody algebras. The central result is that every quantum symmetric subalgebra is realized via a short star product on a quantum horospherical subalgebra by a dual quantum Popov degeneration. The fact that Popov's degenerations are always short star products gives short and conceptual proofs of many results for quantum symmetric pairs that before were proved by long and diverse methods. Our methods prove that all quasi K-matrices for these quantum Kac-Moody symmetric pairs are explicitly expressible in terms of the much studied quasi R-matrices of Drinfeld and Lusztig. This is a joint work with Stefan Kolb.

Program: Titles and Abstracts of Contributed Talks

Juan Camilo Arias

Affiliation: Universidad del Rosario
Email: juancamil.arias@urosario.edu.co
Title: Kashiwara algebra in type $A_2^{(2)}$ and imaginary Verma modules.
Abstract: In this talk, I will present the construction and main properties of the Kashiwara algebra in type $A_2^{(2)}$, which is a essential ingredient in the construction of crytal-like bases for imaginary Verma modules. This is done by defining suitable Kashiwara operators which satisfy analogue properties as the classical ones. This is a joint work with K. C. Misra and V. Futorny.

Devjani Basu

Affiliation: University of Minnesota
Email: devjanib@d.umn.edu
Title: Modular Representations of Special Linear Groups over a Finite Field
Abstract: Modular representations of a finite group are representations over the field F with nonzero characteristic p, where p divides the order of the group. Let $q = p^r$, then the special linear group $SL_n(F_q)$ is a finite group of Lie type. This goal of this talk is to provide an explicit description of the modular representations of this group in defining the characteristic p. We employ some classical methods, such as the classification of simple modules by their highest weight, and we explore some new techniques to accomplish this goal.

Christopher Bendel

Affiliation: University of Wisconsin-Stout
Email: bendelc@uwstout.edu
Title: On Conjectures of Donkin (in Type A)
Abstract: This talk will review the current status of work on two conjectures of Donkin (as well as related conjectures) for representations of a simple algebraic group over a field of prime characteristic: the Tilting Module Conjecture (related to the lifting of structures from a Frobenius kernel) and the good p-filtration conjecture (on the relation between certain filtrations and tensoring with the Steinberg module). The talk will include some recent evidence for validity of the conjectures in Type A.

Ziming Chen

Affiliation: The University of Virginia
Email: jrd6an@virginia.edu
Title: iCanonical basis arising from quasi-split rank one iquantum group
Abstract: This talk concerns the icanonical basis for the quasi-split rank-one iquantum group. After briefly explaining how iquantum groups arise from quantum symmetric pairs and how they extend the usual theory of quantum groups, we first will study simple modules over quantum $\mathfrak{sl}_3$. On these modules, we describe explicit transition formulas among the icanonical basis, a

monomial basis, and the canonical basis. We then explain how the module-level formulas pass, via an asymptotic limit, to the modified iquantum group, yielding analogous transition formulas at the algebra level.

Christopher Drupieski

Affiliation: DePaul University
Email: `c.drupieski@depaul.edu`
Title: Some Lie (super)algebras generated by reflections
Abstract: In 2007, motivated by questions from the representation theory of the braid group, Ivan Marin computed the structure of the Lie algebra generated by the transpositions in the group algebra of the symmetric group. In this talk I'll describe joint work with Jonathan Kujawa investigating analogous questions for arbitrary finite reflection groups, as well as results on the Lie superalgebras generated by the same sets of elements.

Nikolay Grantcharov

Affiliation: University of Georgia
Email: `nikolayg@uga.edu`
Title: Symmetries of cotangent bundles of generalized base affine spaces.
Abstract: Given an affine algebraic variety X over $\mathbb{C}$, a natural question to ask is what symmetries its cotangent bundle admits. One fundamental source of such symmetries is the Fourier transform. We will explain how these Fourier transforms can be adapted to produce symmetries of the affinization of T^*X in the case of the base affine space $X = SL_n/U$ and more generally the case of $X = SL_n/[P, P]$ for P a standard parabolic subgroup. As a byproduct, we construct many examples of affine algebraic varieties that are not isomorphic, but whose cotangent bundles are isomorphic.

Dennis Hou

Affiliation: Rutgers University–New Brunswick
Email: `dhou@rutgers.edu`
Title: Basic and strange Lie superalgebras over noncommutative rings
Abstract: We review the theory of Lie algebras over noncommutative rings (Kapranov 1998; Berenstein–Retakh 2008), by which we mean associative algebras in characteristic 0. We extend this theory to Lie superalgebras of basic and strange type, presenting explicit realizations for each family. Special attention is given to $D(2, 1; \alpha)$, including its real and rational forms.

Naihuan Jing

Affiliation: North Carolina State University
Email: `jing@ncsu.edu`
Title: Quantum Littlewood correspondences
Abstract: In the 1940s Littlewood formulated three fundamental correspondences parallel to the Schur-Weyl duality. We introduce the notion of quantum immanants and establish the quantum

Littlewood correspondences between quantum immanants and Schur functions for the quantum general linear group and the Heck algebra. Via the correspondences, we have found an exact relationship between the Gelfand-Tsetlin bases of $U_q(gl(n))$ and Young's orthonormal bases for the Hecke algebra. This leads to a trace formula for the quantum immanants that has settled the generalization problem of q-analog of Kostant's formular for λ-immanants. As applications, we also derive general q-Littlewood-Merris-Watkins identities and q-Goulden-Jackson identities as special cases of the quantum Littlewood correspondence III. This is joint work with Jian Zhang and Yinlong Liu.

Elizabeth Jurisich

Affiliation: College of Charleston
Email: `JurisichE@cofc.edu`
Title: A Magnus group construction for certain Borcherds algebras
Abstract: Under given conditions on the root system of the Borcherds algebra a Kac-Moody-like group can be constructed as the semi direct product of a Magnus group and the Lie or Kac-Moody group of a semi-simple or Kac-Moody subalgebra.

Scott Larson

Affiliation: University of Georgia
Email: `scott.larson@uga.edu`
Title: Functions on nilpotent orbit covers and birational geometry
Abstract: We use an analogue of the Springer resolution to describe the G-module structure on the ring of regular functions on any cover $\tilde{O}$ of any nilpotent orbit for $G = SL_n$. Building on previous work on the extended Springer resolution, we construct a variety $\tilde{M}$ that is finite over the cotangent bundle of a partial flag variety G/P, and proper and birational over the affinization M of $\tilde{O}$. We use techniques in birational geometry to show that $\tilde{M}$ has rational singularities, which provides us with the necessary cohomology vanishing results to describe the ring of functions on $\tilde{O}$ as an induced representation from a Levi subgroup of G.

Tejbir Lohan

Affiliation: Indian Statistical Institute, Delhi Centre, India
Email: `tejbirlohan70@gmail.com`
Title: Real adjoint orbits of linear Lie groups
Abstract: The study of adjoint orbits in semisimple Lie groups is a central topic in Lie theory, geometry, and representation theory. Let G be a Lie group with Lie algebra $\mathfrak{g}$, and consider the adjoint orbit $\{\mathrm{Ad}(g)X \mid g \in G\}$ of an element $X \in \mathfrak{g}$ under the adjoint representation $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$. An element $X \in \mathfrak{g}$ is called Ad_G-*real* if there exists $g \in G$ such that $\mathrm{Ad}(g)X = -X$, and *strongly* Ad_G-real if this holds for an involution $g \in G$. The classification of Ad_G-real and strongly Ad_G-real elements in a Lie algebra $\mathfrak{g}$ is a problem of broad interest in the study of symmetry, reversibility, matrix factorizations, and representation theory.

In this talk, we present a complete classification of Ad_G-real and strongly Ad_G-real elements in the special linear Lie algebra $\mathfrak{sl}(n, \mathbb{C})$ and the symplectic Lie algebra $\mathfrak{sp}(2n, \mathbb{C})$, and we outline

ongoing work toward analogous results for the complex orthogonal Lie algebra $\mathfrak{so}(n, \mathbb{C})$. This talk is based on joint work with Krishnendu Gongopadhyay and Chandan Maity.

Kang Lu

Affiliation: Southern University of Science and Technology
Email: `kang.lu.math@gmail.com`
Title: iYangians as coideal subalgebras of Yangians
Abstract: In recent joint work with Weiqiang Wang and Weinan Zhang, we established Drinfeld-type current presentations for twisted Yangians of type AI and beyond, which we refer to as iYangians. These presentations are realized via Gauss decomposition and the degeneration of affine iquantum groups. In this talk, I will discuss how to realize these iYangians as coideal subalgebras of standard Yangians utilizing a minimalistic presentation.

Chandan Maity

Affiliation: Indian Institute of Science Education and Research, Berhampur
Email: `cmaity@iiserbpr.ac.in`
Title: Lower dimensional Betti numbers of homogeneous spaces of Lie groups
Abstract: The study of the de Rham cohomology of a homogeneous space G/H, where G is a connected Lie group and H is a closed subgroup, is a classical theme in differential geometry and topology.

In this talk, I will present explicit descriptions of certain low-dimensional Betti numbers of homogeneous spaces of Lie groups. As an application, we obtain an interesting invariant of compact homogeneous spaces expressed in terms of the numbers of simple factors of the ambient group and of the associated closed subgroup. From a computational point of view, these descriptions are new and very useful, and they play an important role in determining the second cohomology of nilpotent orbits; the talk is based on joint work with Indranil Biswas and Pralay Chatterjee.

Ruben Mamani

Affiliation: University of Toledo
Email: `rmamani@rockets.utoledo.edu`
Title: Center and derivations of generalized Weyl algebras over $\mathbb{Z}/p^n\mathbb{Z}$.
Abstract: Let A be either a classical generalized Weyl algebra or $U(\mathfrak{sl}_2)$ over $\mathbb{Z}/p^n\mathbb{Z}$. On this talk we show that the center of $U(\mathfrak{sl}_2)$ is generated by the Casimir element over ring of the Witt vectors (of length $n-1$) of its p-center. Our description of derivations of A implies that the restriction homomorphism $HH^1(A) \to \mathrm{Der}_K(Z(A), Z(A))$ from the space of outer derivations onto the center of A is an isomorphism, for $k = \mathbb{Z}/p\mathbb{Z}$.

Nham Ngo

Affiliation: University of North Georgia
Email: `nvngo@ung.edu`
Title: Linear Bounds for Cohomology of Algebraic Groups

Abstract: In this talk, we present explicit linear bounds for the dimension of the rational cohomology for a simple algebraic group over an algebraically closed field of prime characteristic. These bounds only depend on cohomological degrees and the rank of the group. Our methods are purely elementary combinatorics. This is joint work with Chris Bendel.

Daniel Orr

Affiliation: Virginia Tech
Email: `dorr@vt.edu`
Title: Stable-limit partially symmetric Macdonald functions and parabolic flag Hilbert schemes
Abstract: We study a canonical isomorphism between two representations of a noncommutative algebra arising from Carlsson and Mellit's proof of the shuffle conjecture. The two representations have vastly different origins. One arises from the K-theory of parabolic flag Hilbert schemes, while the other is defined by explicit algebraic operators on a space of functions. Our main result identifies the images of a distinguished geometric basis under this isomorphism, extending Haiman's geometric realization of Macdonald symmetric functions.

Cornelius Pillen

Affiliation: University of South Alabama
Email: `pillen@southalabama.edu`
Title: On a Theorem by Chastkofsky and Jantzen and the Induction Functor
Abstract: Attached to an algebraic group G defined over an algebraically closed field k of characteristic p are two families of group schemes, the r-th Frobenius kernels, denoted by G_r, and the fixed points of iterated Frobenius map, the finite groups of Lie type, denoted by $G(\mathbb{F}_q)$.

In 1981 Chastkofsky and Jantzen, independently, found a formula that describes the decomposition of the Brauer character of a principal indecomposable G_r-module into principal indecomposable modules for $G(\mathbb{F}_q)$. The multiplicities are given in terms of data coming from the algebraic group G. The theorem has found many applications for the finite group $G(\mathbb{F}_q)$, including characters data, cohomology, Cartan invariants and the decomposition of Deligne-Lusztig characters.

Twenty years later Bendel, Nakano and the presenter studied an induction functor from $G(\mathbb{F}_q)$-modules to G-modules. Using filtrations and truncation, large amounts of data coming from the algebraic group can be transferred to the finite group. This talk looks at connections between the theorem of Chastkofsky and Jantzen and the induction functor.

Colton Sandvik

Affiliation: Louisiana State University
Email: `csandv1@lsu.edu`
Title: A Betti geometric Casselman-Shalika equivalence
Abstract: The geometric Casselman-Shalika equivalence of Bezrukavnikov-Gaitsgory-Mirkovic-Riche-Rider gives an equivalence of categories between representations of a reductive group and Iwahori-Whittaker sheaves on the affine Grassmannian for the Langlands dual group. In order to define the automorphic side, one has to restrict the sheaf-theoretic setting to étale sheaves or D-modules so that an exponential local system exists. In this talk, we will discuss a different

incarnation of the Whittaker model called the Kirillov model which makes sense in any sheaf theory. We will then explain how the Kirillov model can be used to develop a Betti version of the geometric Casselman-Shalika equivalence.

Xin Tang

Affiliation: Fayetteville State University
Email: `xtang@uncfsu.edu`
Title: Relative Dixmier Property for Poisson Algebras
Abstract: In this talk, we will present some results concerning the Dixmier property for some families of (Poisson) algebras. One key ingredient to our method is the so-called "relative Dixmier property", which has applications to other topics such as the cancellation problem and non-existence of Hopf coactions.

Aparna Upadhyay

Affiliation: University of South Alabama
Email: `aupadhyay@southalabama.edu`
Title: A Character polynomial for the twisted Foulkes modules
Abstract: The permutation module given by the action of the symmetric group of even degree 2m on the collection of set partitions of a set of size 2m into m sets each of size two is called a Foulkes module. The twisted Foulkes modules are obtained by considering certain twists of the Foulkes module by the sign character. In this talk, we will see a combinatorial description of the ordinary character of the twisted Foulkes modules. We will explore some very interesting interpretations of the twisted Foulkes characters in terms of a character polynomial that encodes all the ordinary characters of the various twisted Foulkes modules of the symmetric group of degree n. This is joint work with Josh Hall.

Kent Vashaw

Affiliation: University of California Los Angeles
Email: `kent.vashaw@gmail.com`
Title: Monoidal triangular geometry for bimodules of unipotent Hopf algebras
Abstract: We describe a program aimed at classifying thick ideals, Balmer spectra, and submodule categories of various stable categories of bimodules and modules for finite dimensional selfinjective algebras, and at clarifying the relationship between the universal Balmer support and the Hochschild cohomology support, focusing mostly on the case of a unipotent Hopf algebra A. The stable left-right projective category of A-bimodules is a monoidal triangulated category under the tensor product over A, and acts naturally on the stable category of A. We construct surjective maps from the Balmer spectrum of the stable category of A to the spectrum of any subcategory of the stable left-right projective category, describe the extent to which this aforementioned spectrum classifies submodule categories via a Stevenson-type theorem, and relate the universal Balmer support to the classical Hochschild cohomology support. This is joint work with Oeyvind Solberg and Sarah Witherspoon.

Trung Vu

Affiliation: Yale university
Email: trung.vu@yale.edu
Title: Quantum Harish-Chandra bimodules at roots of unity
Abstract: Harish-Chandra bimodules were firstly studied by Harish-Chandra in his works about infinite dimensional representations of complex groups as real groups. They are related to many other objects in representation theory and related areas such as category $\mathcal{O}$, character sheaves, knot homology and such. The notation of Harish-Chandra bimodules has been generalized to the context of filtered quantizations, affine Lie algebras, Deligne categories. In this talk, I will introduce the category of Harish-Chandra bimodules for quantum groups. When the parameter q of quantum group is an odd order roots of unity, I will relate the categories of quantum Harish-Chandra bimodules to the category of affine Soergel bimodules and Non-commutative Springer resolutions.

Arik Wilbert

Affiliation: University of South Alabama
Email: wilbert@southalabama.edu
Title: Category $\mathcal{O}$ for Lie superalgebras
Abstract: In this talk, we introduce a Category $\mathcal{O}$ for any quasi-reductive Lie superalgebra $\mathfrak{g}$ with respect to a triangular decomposition. When the decomposition arises from a principal parabolic subalgebra $\mathfrak{p}$ of $\mathfrak{g}$, the Category $\mathcal{O}$ exhibits rich homological properties. For one, the Category $\mathcal{O}$ is standardly stratified. Furthermore, the categorical cohomology of $\mathcal{O}$ is a finitely generated ring. Finally, the complexity of modules in Category $\mathcal{O}$ is finite with an explicit upper bound given by the dimension of the subspace of the odd degree elements in $\mathfrak{g}$. This latter result generalizes work by Coulembier–Serganova for $\mathfrak{gl}(m|n)$. This is joint work with C.-J. Lai and D.K. Nakano.

Haihan Wu

Affiliation: Johns Hopkins University
Email: hwu125@jh.edu
Title: Webs and pockets in SL(4) affine building
Abstract: To study the representation theory of quantum SL(2) diagrammatically, one can study the Temperley-Lieb category, whose basis morphisms are planar matchings. A generalization to the SL(3) case gives rise to the SL(3) web category, whose basis morphisms are given by non-elliptic webs - planar trivalent graphs with girth greater or equal to 6. It is shown by Fontaine-Kamnitzer-Kuperberg that the dual graphs of SL(3) basis webs are embedded isometrically into the SL(3) affine building, which is defined as a simplicial complex with vertices given by elements in the affine Grassmannian. In this talk, I will review background materials and talk about a generalization of the embeddings to the SL(4) case, based on joint work (in progress) with Gaetz-Striker-Swanson.

Ziqing Xiang

Affiliation: Southern University of Science and Technology

Email: xiangzq@sustech.edu.cn
Title: Wreath modules for quantum wreath product
Abstract: Quantum wreath products provide a unified framework to study many algebras, such as Ariki-Koike algebras and affine Hecke algebras. In this talk, we will discuss ways to construct modules for these algebras, called wreath modules, and show that many interesting modules can be obtained in this way.

Kaitao Xie

Affiliation: The University of Hong Kong
Email: kaitaoxie@connect.hku.hk
Title: Total Positivity and Remarkable Polyhedral Spaces
Abstract: As a topic in Lie theory, total positivity has found connections to such diverse areas as combinatorics, quantum group, cluster algebra, higher Teichmuller theory, and theoretical physics. Many topological objects in total positivity appear to share some remarkable properties. In this talk, we explore these properties for the twisted flag varieties, double flag varieties, double Bruhat cells of Kac-Moody groups, and the wonderful compactifications of semisimple groups. We also present a conjecture about canonical basis in quantum group. This is based on some recent joint works with Xuhua He.

Jize Yu

Affiliation: Georgia Southern University
Email: jize.yu.math@gmail.com
Title: Tamely ramifield local relative Langlands duality via categorical representation theory
Abstract: Relative Langlands duality, as formulated by Ben-Zvi, Sakellaridis, and Venkatesh, predicts that the harmonic analysis of a spherical G-variety X is governed by a Hamiltonian space $M^\vee$ for the Langlands dual group $G^\vee$. Its local categorical form relates sheaves on the loop space of X to coherent sheaves on $M^\vee$. In this talk, I will discuss a tamely ramified, or Iwahori-level, version of this correspondence, following Devalapurkar's conjecture. In the group case $G = H \times H$, $X = H$, this conjecture closely relates to the tamely ramified local geometric Langlands correspondence established by Bezrukavnikov. Under natural hypotheses, we prove Devalapurkar's conjecture for a broad class of spherical varieties. The key ingredient of the proof uses categorical representation theory to pass from spherical to Iwahori level. This is joint work with Milton Lin and Toan Pham.

Fan Zhou

Affiliation: Columbia University
Email: fz2326@columbia.edu
Title: Categorifying Jacobi-Trudi
Abstract: We discuss a way to categorify the famous Jacobi-Trudi identity by constructing a quasi-hereditary algebra A with a map kS_n to A; the "dominant simple" of A admits a BGG resolution, and after restricting to kS_n, we obtain a resolution of a simple module of kS_n by permutation modules. This is proven using Koszul theory – we show that A has a 'nil-algebra' which is Koszul, and the BGG resolution is recovered from Koszul duality with respect to this

nil-algebra. Interestingly, this story works in positive characteristic ($p > l(\lambda)$) as well.

Songhao Zhu

Affiliation: Georgia Institute of Technology
Email: `szhu95@gatech.edu`
Title: Super Multivariate Krawtchouk Polynomials via Lie Superalgebras
Abstract: Krawtchouk polynomials constitute a family of orthogonal polynomials with applications in probability theory. In 2012, Iliev gave a Lie-theoretic interpretation of the multivariate Krawtchouk polynomials using certain representations of the Lie algebra $\mathfrak{sl}_n$. In this talk, we will present a superization of this interpretation, introduce super Krawtchouk polynomials using anticommuting variables, and explore their connection with zonal spherical functions on oriented Grassmannians. This is a joint work in progress with Plamen Iliev.